



Panchip Microelectronics Co., Ltd.

PAN3029/3060 Series Intelligent Search Application Reference Instructions

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Documentation

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Revision History

Version	Revision date	Description
V1.0	2023.06	Create initial version
V1.1	2024.03	Added smart search description
V1.2	2024.04	Modify smart search description

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1 Introduction

The PAN3029/3060 series chips provide intelligent search functions to realize lightweight gateway devices. They can intelligently identify the SF parameters in the channel during reception, so as to receive different SF signal data.

2 Software Design Reference

When using this function, you need to perform corresponding configurations on the transmitter and receiver respectively. **Notes:** Because the SDK will automatically turn on LDR for low-rate parameters, in order to ensure that both the sender and the receiver use the same LDR configuration, both the sender and the receiver need to turn off LDR when using this function.

SDK interface function:

```
int calculate_chirp_count(int sf_range[], int size, int chirp_counts[]);  
RF_Err_t rf_set_auto_sf_tx_preamble(int sf, int sf_range[], int size, int chirp_counts[]);  
RF_Err_t rf_set_auto_sf_rx_on(int sf_range[], int size);  
RF_Err_t rf_set_auto_sf_rx_off(void);
```

Parameters:

- sf_range[]: SF values supported by smart search
- Size: The number of SFs corresponding to sf_range
- chirp_counts[]: The number of leading codes corresponding to the SF value supported by smart search
- Sf: The SF value transmitted by the TX side

2.1 Receiver Configuration

Before receiving, configure `rf_set_auto_sf_rx_on(sf_range, size)` on the receiving end to turn on the smart search function. At this time, the chip can receive data from different SF signals on the same channel, and then receive it according to the normal process.

To disable this function, configure `rf_set_auto_sf_rx_off()` on the receiving side. At this time, the chip can only receive data from the specified SF signal on the same channel, and then receive it according to the normal process.

2.2 Transmitter Configuration

The sender needs to call `calculate_chirp_count(sf_range, size, chirp_counts)` before sending to calculate the number of preamble codes corresponding to the SF value supported by smart search, and then configure `rf_set_auto_sf_tx_preamble(DEFAULT_SF, sf_range, size, chirp_counts)`. The function will configure different preamble lengths according to different DEFAULT_SF values.

In the smart search mode, the preamble length of the transmitter will be appropriately increased, which will increase the chip's transmission duration. To disable this function, the transmitter needs to configure `rf_set_auto_sf_rx_off()`, and the function will restore the preamble value to the default value.

2.3 Intelligent search range configuration

The smart search function supports configuring different SF search ranges according to actual needs. It is recommended to select continuous SF values for smart search, such as SF7~9.

2.4 Notes

The SDK will automatically turn on LDR for low-rate parameters. To ensure that both the sender and receiver use the same LDR configuration, both sender and receiver need to turn off LDR when using the smart search function.